Hugo Marquerie 12/01/2026

Fórmula del núcleo de Dirichlet

Lema 1. *Sea* $\{D_N\}_{N\in\mathbb{N}}$ *el núcleo de Dirichlet, entonces*

(1) $\forall N \in \mathbb{N} : D_N$ *es* 1*-periódica.*

(2) $\forall N \in \mathbb{N}, x \in \left[-1/2, 1/2\right) : D_N(x) = \dfrac{\sin\left((2N+1)\pi x\right)}{\sin(\pi x)}$ *si* $x \neq 0$ *y* $D_N(0) = 2N+1$.

***Demostración*:**

(1) Como $D_N(x)$ es una suma finita de funciones 1-periódicas, es 1-periódica.

(2) Si $x = 0$, entonces $D_N(0) = \sum_{n=-N}^{N} 1 = 2N+1$. Si $x \neq 0$, entonces

$$\begin{aligned}\forall N \in \mathbb{N} : D_N(x) = \sum_{n=-N}^{N} e^{2\pi i n x} &= \frac{e^{2\pi i N x}e^{2\pi i x} - e^{-2\pi i N x}}{e^{2\pi i x} - 1} \\ &= \frac{e^{\pi i x}}{e^{\pi i x}} \cdot \frac{e^{\pi i (2N+1)x} - e^{-\pi i (2N+1)x}}{e^{\pi i x} - e^{-\pi i x}} = \frac{\sin\left((2N+1)\pi x\right)}{\sin(\pi x)}.\end{aligned}$$

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Referenciado en

- `prop-criterio-dirichlet`
- `prop-criterio-dini`
- `lem-nucleo-dirichlet-no-es-nucleo-sumabilidad`
- `lem-formula-nucleo-fejer`

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